EXPERIMENT – 16

SERIAL INTERRUPT USING UART (KEYBOARD INPUT) IN KEIL

AIM: To write and simulate an Embedded C program using serial interrupt in the 8051 microcontroller so that a keyboard input generates an interrupt and is displayed through UART.

OBJECTIVES:

1. To configure UART in Mode 1 at 9600 baud rate.
2. To enable and service the serial interrupt using an ISR.
3. To display the received keyboard character through UART.

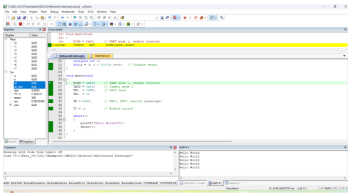
PROGRAM:

```
#include <reg51.h>
#include <stdio.h>
void serial_ISR(void) interrupt 4
{
    char ch;
    if(RI)
    {
        RI = 0;
        ch = SBUF;
        printf("\n>>> INTERRUPT RECEIVED: ");
        printf("%c\n", ch);
    }
}
void delay(void)
{
    unsigned int i;
    for(i=0;i<50000;i++);
}
void main(void)
{
    SCON = 0x50;
    TMOD = 0x20;
    TH1 = 0xFD;
    TR1 = 1;
    IE = 0x90;
    TI = 1;
    while(1)
    {
        printf("Hello World\n");
        delay();
    }
}
```

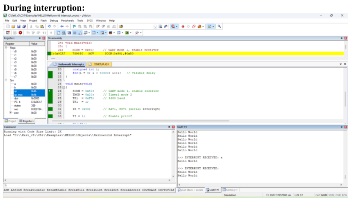
OUTPUT IMAGE:

OUTPUT:

a) Displaying a Message:



b) During interruption:



OBSERVATION:

Action	Output in Serial Window
Program running	Hello World printed repeatedly
Key pressed (e.g., A)	>>> INTERRUPT RECEIVED: A

RESULT:

Thus, the serial interrupt using UART was successfully implemented and verified in Keil simulator. The interrupt was triggered upon keyboard input and serviced correctly using an Interrupt Service Routine.